Answer Key - Mathematics - K-12 Level Assessment 4 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) 10; B) 100; C) 1,000; D) 10,000
   Note: Correct option: A - 10

2. **Answer:** B
   Options: A) (5 x 4) x (6 x 5); B) (5 x 5) + (5 x 4); C) (5 x 5) + (5 x 9); D) (5 x 9) x (6 x 9)
   Note: Correct option: B - (5 x 5) + (5 x 4)

3. **Answer:** C
   Options: A) Small sample size and 95% confidence; B) Small sample size and 99% confidence; C) Large sample size and 95% confidence; D) Large sample size and 99% confidence
   Note: Correct option: C - Large sample size and 95% confidence

4. **Answer:** C
   Options: A) 4; B) 6; C) 10; D) 15
   Note: Correct option: C - 10

5. **Answer:** A
   Options: A) 6; B) 9; C) 12; D) 14
   Note: Correct option: A - 6

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '$52.80'.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '$21.88'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '48'.

Long Form Response

11. **Answer:** We rewrite the given equation as

$$5^{a_{n+1}-a_n} = 1 + \frac{1}{n + \frac{2}{3}} = \frac{3n+5}{3n+2}.$$

Then, we observe a telescoping product:

$$5^{a_n - a_1} = 5^{a_2 - a_1} \cdot 5^{a_3 - a_2} \cdots 5^{a_n - a_{n-1}} = \frac{8}{5} \cdot \frac{11}{8} \cdots \frac{3n+2}{3n-1} = \frac{3n+2}{5}.$$

Since $a_1 = 1$, we have

$$5^{a_n} = 3n + 2$$

for all $n \geq 1$. Thus, a_k is an integer if and only if $3k + 2$ is a power of 5. The next power of 5 which is of the form $3k + 2$ is $5^3 = 125$, which is $3(41) + 2$. Thus $k = 41$.
Note: Students should show all steps in their mathematical solution.

12. **Answer:** Divide 30 by 5 to find 6 teams.. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key